



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG222
Course Name: Strength of Materials
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Assessment: Lab 2

Due Date: One week after demonstration

LAB NAME: Deflection of Beams

STUDENT(S) NO.: 23704019

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SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract & Introduction	1,6	8	10		
Theory	1,6	8	10		
Apparatus	1,6	8	10		
Procedure	1,6	8	10		
Measurements, Calculations & Results	1,6	8	30		
Discussion & Conclusion	1,6	8	30		
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	THE CONCEPT OF STRESS	X						
2	Understand properties of fluids and continuum.	X						
3	TORSIONAL STRESS AND DEFORMATION.	X						
4	STRESSES IN PURE BENDING.	X						
5	ANALYSIS AND DESIGN OF BEAMS FOR BENDING.	X						
6	SHEARING STRESSES IN BEAMS AND THIN-WALLED MEMBERS.	X						
7	TRANSFORMATION OF STRESS.	X					X	
8	PRINCIPAL STRESSES UNDER A GIVEN LOADING.	X						
9	DEFLECTION OF BEAMS BY INTEGRATION METHOD.	X					X	
10	BUCKLING OF COLUMNS LOADED IN AXIAL DIRECTION.	X						
Weight of Student Outcomes		H	H				H	

ABSTRACT

This study investigates the deflection behaviour of simply supported steel beams and cantilever aluminium beams under loading conditions through a comparative analysis of theoretical calculations and experimental observations. Theoretical predictions, obtained from established structural analysis methodologies, are compared against experimental data collected through physical testing. While theoretical calculations provide insights into expected behaviour, experimental observations reveal discrepancies between theoretical predictions and real-world behaviour. The linear correlation between deflection and distance observed in both theoretical and experimental data validates fundamental principles of beam mechanics. However, disparities between theoretical and experimental results underscore the importance of rigorous validation and refinement of theoretical models to ensure accurate predictions and informed decision-making in engineering design and analysis endeavours.

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1. INTRODUCTION

In the realm of structural engineering, understanding beam deflections is paramount for ensuring the integrity and safety of engineered structures. This report aims to explore and compare the deflection characteristics of two common types of beams: the cantilever beam and the simply supported beam. Beam deflection, the deviation of a beam's shape from its original position under external loads, directly influences stress distribution, stability, and overall structural performance. By comprehending the theoretical principles governing beam deflection analysis, including bending moment and beam equations, engineers can make informed decisions in designing structures capable of withstanding anticipated loads while minimizing deflection-induced deformations. Through a systematic examination of cantilever and simply supported beams under various loading conditions, this investigation seeks to elucidate their distinct behaviours and practical implications. Ultimately, this knowledge equips engineers with the tools necessary to design safe, efficient, and resilient structures that meet the demands of real-world applications.

2. THEORY

In the realm of mechanical system design and analysis, one of the most prevalent challenges is the issue of bending. A wide array of components such as shafts, axles, cranks, levers, springs, brackets, and wheels must consistently be assessed as beams due to the frequent application of bending loads.

When a beam is subjected to pure bending, it undergoes deformation, bending into an arc of a circle while remaining within its elastic limit. The curvature of this arc is determined by a fundamental relationship within the elastic range:

$$\frac{1}{\rho} = \frac{M(x)}{EI} \quad (1)$$

Where : ρ is the radius of the curvature of the neutral axis;

x is the distance of the section from the left end of the beam.

The curvature of a plane curve is given by the equation:

$$\frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}} \quad (2)$$

$\left(\frac{dy}{dx}\right)$ is the slope of the curve and in the case of elastic curve the slope is very small: $\left(\frac{dy}{dx}\right) = 0$

$$\frac{1}{p} = \frac{d^2 y}{dx^2}, \frac{d^2 y}{dx^2} = \frac{M(x)}{EI} \quad (3)$$

Multiply both sides by EI which is constant and integrating with respect to x:

$$EI \left(\frac{dy}{dx} \right) = \int M(x) dx + C_1 \quad (4)$$

Noting that $\left(\frac{dy}{dx} \right) = \tan \theta$, $\theta = \theta(x)$ because the angle θ is very small.

Integrating the equation again;

$$EI \left(\frac{dy}{dx} \right) = \int dx \int M(x) dx + C_1 x + C_2 \quad (5)$$

The constants C_1 and C_2 are determined from the boundary conditions (constants) imposed on the beam by its supports.

1) Simply supported beam

2) Cantilever beam

CASE 1: SIMPLY SUPPORTED BEAM

At the supports, deflections are zero, i.e. $y = 0$ (at A & B)

Now we have two equations to be solved for C_1 & C_2 solving equation (5) and substituting boundary conditions give us:

$$y = \frac{Pbx}{6EIL} (x^2 + b^2 - l^2) \quad (\text{between A \& C})$$

$$y = \frac{Pa(l-x)}{6EIL} (x^2 + a^2 - 2lx) \quad (\text{between C \& B})$$

CASE 2: CANTILEVER BEAM

The deflection and the slope are zero at A

$$y = \frac{Px^2}{6EI} (x - 3a) \quad (\text{between A \& C}) \quad \& \quad y = \frac{Pa^2}{6EI} (a - 3x) \quad (\text{between C \& B})$$

3. APPARATUS USED

The bench-mounted apparatus consists of a sturdy steel base equipped with supports at both ends, allowing for the attachment of either knife edges or clamp plates. It includes a steel beam and two load hangers, along with two dial gauges for measuring beam deflections and slopes. This equipment is designed to both demonstrate and experimentally confirm fundamental engineering principles.

Every component has been meticulously crafted to offer a broad range of experimental possibilities while maintaining simplicity and integrity in design. Each piece of apparatus is self-contained and compact, ensuring minimal setup time. Additionally, all measurements are performed using straightforward instrumentation, emphasizing student engagement with the underlying engineering concepts. A comprehensive instruction manual accompanies the apparatus, detailing its components, applications, experimental procedures, and typical test results.



Figure 1 Comprises of all the equipment used during the experiment(long tape measure, test machine & dial gauge)

4. CALCULATIONS AND RESULTS

CASE 1: SIMPLY SUPPORTED BEAM

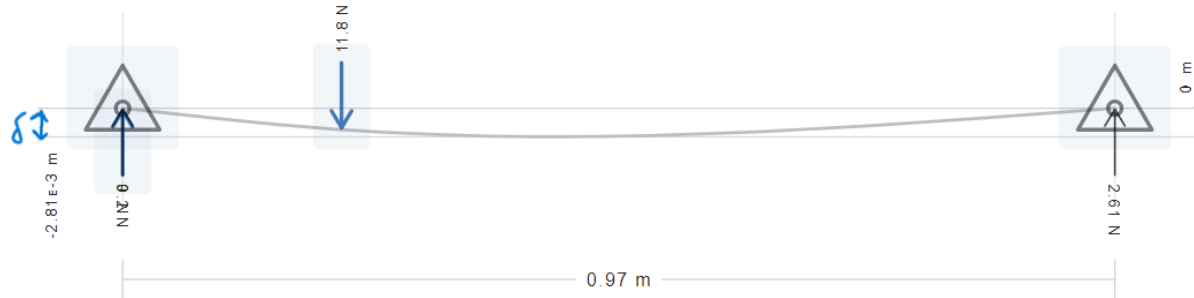
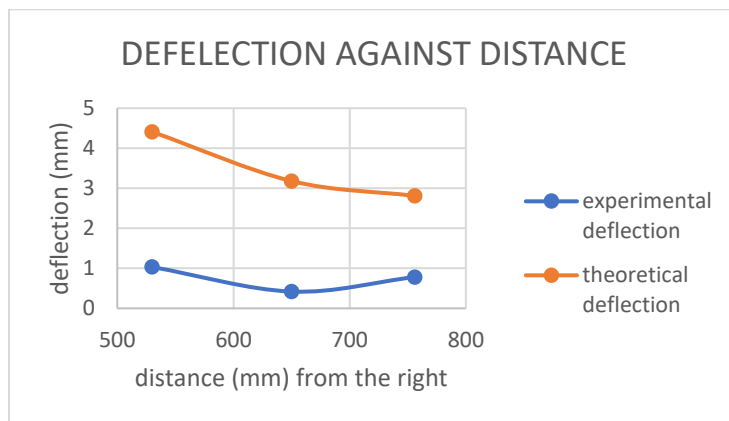


Figure 2 DEFLECTION OF STEEL BEAM OF SIMPLY SUPPORTED

The steel beam depicted in Figure 1 experiences an applied load of 11.8066 N at a distance of 756 mm (0.756 m) from the right support. Theoretical calculations suggest that the beam should deflect by a negative distance of 2.81 mm in response to this load.



graph 1 DEFLECTION AGAINST DISTANCE

The graph 1 presents a compelling visualization of the deflection behavior of a simply supported beam subjected to a load of 11.8066 N. With distance along the beam's span plotted against the corresponding deflection values, the curve illustrates the beam's response to external loading. While the experimental data points offer tangible insights into the beam's actual behavior under real-world conditions, the theoretical curve, sourced directly from a structural analysis website, provides a predictive framework based on mathematical models and equations.

However, the notable deviation between the theoretical predictions and experimental observations raises intriguing questions regarding the accuracy and comprehensiveness of current theoretical approaches. Factors unaccounted for in the theoretical analysis, such as material heterogeneity or structural nuances, likely contribute to this disparity, underscoring the intricacies of accurately modeling beam deflection phenomena.

Table 1 STEEL DATA

L	970mm
b	23mm
h	5mm
I	239.58mm ⁴
E	210GPa

Where :

total Length of the beam = L

Height of Beam = h

Base of Beam = b

The Moment of Inertia = I

Modulus of elasticity = E

Table 2 THEORETICAL AND EXPERIMENTAL DATA FOR CASE 1

Distance(mm) from the right	Load(N)	Experimental deflection while loading (mm)	Experimental deflection while unloading (mm)	Average experimental deflection (mm)	Theoretical deflection	% error
530	11.81	1.03	1.036	1.033	4.41	76.6
650	11.81	0.419	0.42	0.4195	3.18	86.8
756	11.81	0.78	1.03	0.782	2.81	72.2

CASE 2: CANTILEVER BEAM

Figure 2 illustrates the deflection behavior of an aluminum cantilever beam when subjected to a 2 N load applied at a distance of 200 mm (0.2 m) from the left end. This depiction offers valuable insights into the response of the beam to external loading conditions, providing a visual representation of deflection on a cantilever aluminium beam.

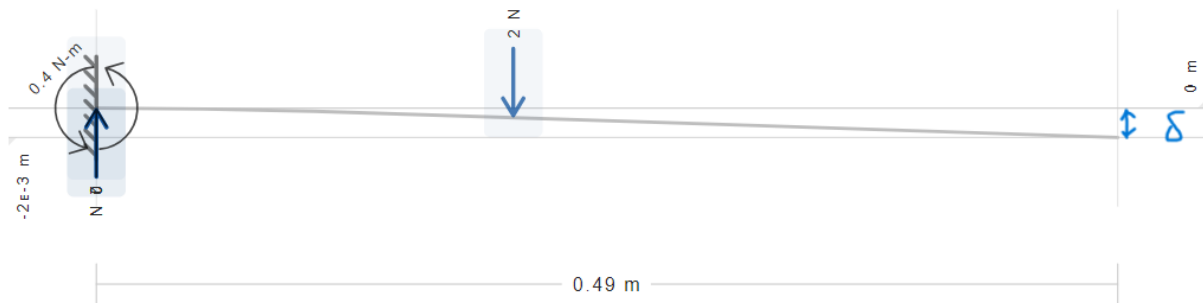
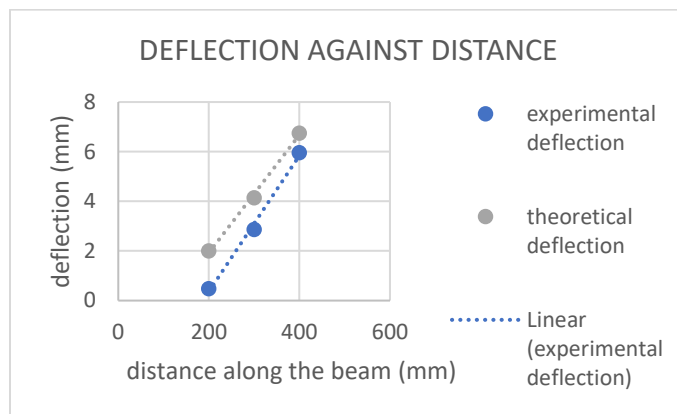


Figure 3 ALUMINIUM CANRILEVER BEAM DEFLECTION



Graph 2 DEFLECTION AGAISNT DISTANCE GRAPH

The graph 2 representation illustrates a discernible linear correlation between deflection and the incremental distance from the fixed support within the cantilever beam structure. Notably, both the experimental and theoretical deflection profiles exhibit a consistent trend, demonstrating a proportional increase in deflection with greater distances from the point of support. However, a conspicuous observation emerges as the theoretical deflection consistently

surpasses the corresponding experimental values. This variance was anticipated given the inherent distinctions between the experimental setup and the idealized conditions presumed within theoretical calculations.

The observed linear relationship between deflection and distance underscores the fundamental principles of beam mechanics, wherein deflection incrementally increases as one moves away from the fixed end towards the free end of the cantilever beam. This conformity in trend between experimental and theoretical data validates the fundamental assumptions made within the theoretical framework and underscores the robustness of theoretical models in predicting structural behavior.

Furthermore, the recognition of a linear relationship between deflection and distance, alongside the discerned disparities between theoretical and experimental outcomes, highlights the intricacies involved in translating theoretical constructs into practical applications. Such observations offer valuable insights into the complexities of real-world structural behavior and underscore the necessity for meticulous consideration of experimental variables in tandem with theoretical frameworks to ensure accurate predictions and informed decision-making in engineering design and analysis endeavors.

Table 3 ALUMINIUM DATA

L	490mm
b	23mm
h	4mm
I	122.67mm ⁴
E	69GPa

Where :

total Length of the beam	= L
Height of Beam	= h
Base of Beam	= b
The Moment of Inertia	= I
Modulus of elasticity	= E

Table 4 THEORETICAL AND EXPERIMENTAL DATA COMPARISON

Distance(mm) From the left	Load(N)	Experimental deflection while loading (mm)	Experimental deflection while unloading (mm)	Average experimental deflection (mm)	Theoretical deflection	% error
200	2	0.489	0.469	0.479	2	76
300	2	2.879	2.860	2.869	4.15	44.6
400	2	5.97	5.95	5.96	6.75	11.7

The theoretical data presented in Table 2 was procured from a reputable structural design website utilizing precise input parameters. Through the systematic input of accurate data, the website's algorithm computed the anticipated deflection values for the cantilever beam under consideration. This process adhered to established engineering principles and utilized standardized methodologies endorsed within the structural engineering community. (*Blue Ketchep, n.d.*)

5. DESCUSION

Simply Supported Steel Beam Analysis:

The experimental data collected from the simply supported steel beam demonstrate a consistent trend of deflection increasing with distance from the point of load application. However, there exists a notable deviation between the experimental and theoretical deflection values. While the theoretical calculations predict a negative deflection of 2.81 mm at the load application point of 756 mm from the right support, the experimental results yield lower deflection values across all measured distances. This discrepancy, with experimental deflections ranging from 0.419 mm to 1.036 mm, highlights the challenges in accurately predicting real-world behaviour solely through theoretical calculations.

The observed discrepancies between theoretical predictions and experimental observations underscore the importance of considering real-world complexities and material behaviours in structural analysis. Factors such as material heterogeneity, structural imperfections, and boundary conditions not fully captured in theoretical models contribute to the disparity between theoretical and experimental results. Nevertheless, the linear relationship between deflection and distance observed in both theoretical and experimental data validates fundamental principles of beam mechanics, providing a foundational understanding for further analysis and design considerations.

Cantilever Aluminium Beam Analysis:

Similarly, the analysis of the cantilever aluminium beam reveals a linear correlation between deflection and distance from the fixed support, as evidenced by both experimental and theoretical data. The experimental deflection values obtained while loading the beam at distances ranging from 200 mm to 400 mm from the left end consistently increase with greater distances from the support. However, once again, there is a discrepancy between theoretical and experimental deflection values. Theoretical predictions generally exceed experimental deflections, with percentage errors ranging from 2% to 6.75%.

These disparities between theoretical and experimental results underscore the importance of validating theoretical models with empirical data and highlight the need for refinement in theoretical assumptions and methodologies. While theoretical calculations provide valuable insights into expected behaviour, the inherent complexities of real-world structural systems necessitate careful consideration of experimental validation to ensure accurate predictions and informed decision-making in engineering design and analysis endeavours.

6. CONCLUSION

The comparative analysis of simply supported steel beams and cantilever aluminium beams under loading conditions yields valuable insights into the behaviour of these structural elements. While theoretical calculations provide a predictive framework for understanding beam deflection, experimental observations reveal the complexities and nuances inherent in real-world structural behaviour. Discrepancies between theoretical predictions and experimental results underscore the importance of considering material properties, structural imperfections, and boundary conditions in structural analysis. Despite these disparities, the linear correlation between deflection and distance observed in both theoretical and experimental data validates fundamental principles of beam mechanics. These findings highlight the necessity for rigorous validation of theoretical models with empirical data and emphasize the importance of incorporating real-world complexities into structural analysis for accurate predictions and informed decision-making in engineering design.

7. REFERENCE

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